ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



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NGHIÊN CỨU GIẢI PHÁP TỐI ƯU CẤU TRÚC CỦA ĐỘI HÌNH ROBOT PHỐI HỢP ĐẨY VẬT

ĐỒ ÁN TỐT NGHIỆP ĐẠI HỌC HỆ CHÍNH QUY

Ngành: Kỹ thuật Robot

HÀ NỘI - 2025

CHƯƠNG 1

SOA and Related Works

1.1 Inertial Parameters Estimation

To the best of our knowledge, there has been no comprehensive review dedicated specifically to center-of-mass (CoM) estimation for manipulated objects. Existing surveys primarily focus on the identification of the complete set of inertial parameters rather than CoM estimation itself. A survey of Schedlinsk [1] presents a literature review of existing methodologies on how to estimate these parameters in mechanical engineering, whereas Mavrakis and Stolkin [2] is specific to robotics. According to Mavrakis and Stolkin's taxonomy, the methods for estimating inertial parameters can be categorized into three main groups by degree of interaction — purely visual, exploratory, and fixed-object.

The first visual methods relied on the assumption of a known and uniform object density in order to simplify the estimation problem by reducing it to the computation of the object's volume. In , [3] the object is epresented as a collection of equally sized voxels with identical masses, from which the center of mass is determined by computing the moment of inertia matrix. One of the most widely recognized approaches is presented in [4], the author decomposed a rigid body into polyhedral components with uniform density and computed the mass, and inertia tensor of each component before combining them to obtain the inertial properties of the complete object. The computation was achieved by converting the required three-dimensional volume integrals into two-dimensional surface integrals and one-dimensional line integrals using the divergence theorem, projection, and Green's theorem. However, while visual perception can accurately capture an object's shape and volume, the underlying density distribution is generally unknown, may vary throughout the object, and cannot be directly observed solely from visual information.

Exploratory and fixed-object methods, on the other hand, involve physical interaction with the object to perform the estimation task. These interactive techniques are commonly formulated based on the Newton–Euler equations of motion, which establish a mathematical relationship between the forces/torques (wrenches) applied to an object and its resulting motion. Properties like the center of mass (CoM), mass and the inertia tensor are treated as constant parameters within these equations and are subsequently estimated from the resulting system of equations, typically using least-squares techniques. In fixed object methods [5, 6, 7], the unknown object is rigidly connected to the robotic system, during the estimation process, either through a firm grasp or by being directly attached to the end-effector. Excitation trajectories are then executed by the robotic system to retrieve the necessary data from wrist sensors. The approach proposed in [8] employed a robotic arm to perform three distinct trajectories to distinguish the individual effects of mass, CoM, and the inertia tensor. The fundamental principle of this method can also be applied to different robotic platforms. In [9], the authors use a UAV equipped with two customised grippers to grasp the object and estimated the payload mass, CoM, and inertia from the UAV dynamics and measured motion, while [10] extended this concept to a hexacopter equipped with a 2-DOF manipulator, enabling simultaneous payload estimation and adaptive motion control. Although this method can achieve the most accurate full three-dimensional inertial properties under controlled conditions, its requirements for a secure grasp and specialized force/torque sensors makes it less versatile than exploratory methods in unconstrained or hazardous scenarios.

Exploratory methods exploit the measurements by utilizing more basic interaction such as such as poking, pushing or tilting. As mentioned, early studies of this category similarly estimates the unknown properties by using the model-based approach. Yu et al. [11] employed a two-fingered robot manipulator equipped with force sensors to estimate an object's mass, 2D CoM, rotational inertia, and motion friction parameters from multiple pushing motions using a least-squares approach. For CoM estimation, they introduced C.M. lines, which are parallel to the object's straight-motion direction and pass through its CoM. These lines were identified from fingertip force and motion measurements under different pushing directions, with their intersection determining the CoM. Franchi et al.[12] developed a decentralized approach for estimating an object's mass, 2D CoM, and rotational inertia using multiple mobile robots. Information on the applied forces and contact geometry is shared among the robots through a consensus algorithm, enabling distributed

estimation of the unknown inertial properties. The method was subsequently extended in [13] to rely solely on noisy end-effector velocity measurements.

Another emerging line of research focuses on CoM estimation without requiring an explicit dynamic model. A number of studies have been built upon Mason's Voting Theorem [14], which establishes a geometric relationship between the pushing direction and the resulting rotational motion of a planar object, and reducing the search space through successive bisection. By additionally incorporating modified binary search algorithm [15], Hyland et al. [16] proposed iterative pushing method using a holonomic mobile robot and external motion capture to derive the CoM location from multiple lines on which the CoM lies, called lines of action (LoA). This works is later extended in [17] into an infrastructure-free approach by utilizing onboard force and vision sensors. Uemura et al. [18] presented a dynamic control based method utilizing a nonlinear feedback control law to continuously adjusting the finger's motion to suppress the object's rotation, achieving pure translational motion along the identified line of action (LoA).

A growing body of work has explored data-driven approaches that learn CoM-related information directly from observed interaction data. Gao et al. [19] introduce a force-sensor-free exploratory framework that integrates high-speed robotic pushing with Mason's Voting Theorem and recurrent neural networks to estimate the CoM of previously unseen objects with non-uniform density. In [20], the authors introduces a data-driven framework that estimates an object's 2D inertial parameters by training a Multi-Output Regression Random Forest on physical features extracted from a single robotic push. By combining analytical pushing mechanics with machine learning, the method achieves robust and accurate localization across various surface types without the need for specialized hardware. These methods enable rapid estimation with fewer physical interactions. However, their reliance on prior training data may limit their generalization to previously unseen objects.

CHƯƠNG 2

Methodology

2.1 Methodology

2.1.1 Formulation

The system consists of N omnidirectional mobile robots manipulating an object of polygonal shape whose motion is constrained to a 2D plane using pushing interactions, with all positions, velocities, and forces expressed in a world-fixed reference frame $\mathcal{W}$. The primary objective is for the robots to cooperatively estimate the center of mass (CoM) of the object in a distributed manner. It is assumed that the object's pose (position and orientation) and 2D geometry are known to all robots and neither the object nor the robots are subject to kinematic constraints. Each robot i has a circular footprint with radius r_R, can exert a normal force f_i within a limited range, $0 \leq f_i \leq f_{\max}$, and measure the position $\mathbf{p}_i$ and velocity $\dot{\mathbf{p}}_i$ at its contact point C_i on the object's surface, as well as the object's angular velocity ω_o. The forces applied to the object by the robots are represented by the vector $\mathbf{f} = [f_1, \ldots, f_N]^T \in \mathcal{F} := \left\{\mathbf{f} \in \mathbb{R}^N \mid 0 \leq f_i \leq f_{\max},\ i = 1, \ldots, N\right\}$. It is also assumed that the robots are able to maintain continuous contact with the object and accurately regulate the pushing force at their corresponding contact points throughout the manipulation process. The net pushing force and torque exerted by all robots on the object are described as:

$$\mathbf{F} = \sum_{i=1}^{N} \mathbf{F}_i \tag{2.1}$$

$$\tau = \sum_{i=1}^{N} \left(\mathbf{p}_i - \mathbf{p}_C\right)^{\perp T} \mathbf{F}_i \tag{2.2}$$

where $\mathbf{p}_C$ denotes the position of the object's CoM, and $(\cdot)^\perp$ denotes the perpendicular operator, defined as $\mathbf{v}^\perp = \mathbf{Q}\mathbf{v}$, with $\mathbf{Q} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. $\mathbf{F}_i = f_i \mathbf{n}_i$ is the force vector of robot i, with $\mathbf{n}_i$ being the inward unit normal vector of the object's boundary at the contact point C_i. Friction between the object and the ground is not explicitly modeled, since an accurate dynamic prediction of the object's motion is not required. Equation (2.2) can be rewritten as:

$$\tau = \sum_{i=1}^{N} \mathbf{p}_i^{\perp T} \mathbf{F}_i - \mathbf{p}_C^{\perp T} \mathbf{F} \tag{2.3}$$

2.1.2 Estimation algorithm

When the object undergoes planar rotational motion, all contact points rotate instantaneously about the same instantaneous center of rotation (ICR). The velocity of each contact point is tangential to its instantaneous circular trajectory and therefore perpendicular to the line connecting the contact point to the ICR. When $\omega_o \neq 0$, the ICR can be geometrically determined from the velocity measurements at any contact point using the following equation:

$$\mathbf{p}_{\text{ICR}} = \mathbf{p}_i + \frac{1}{\omega_o} (\dot{\mathbf{p}}_i)^\perp \tag{2.4}$$

The proposed distributed estimation strategy aims to generate approximately pure rotational motion of the object by coordinating the robot formation and the forces exerted by the robots such that the net force on the object vanishes, i.e., $\mathbf{F} = \mathbf{0}$, while generating a nonzero nominal desired torque τ_d. This condition decouples the net torque from the CoM position and reduces (2.3) to:

$$\tau = \sum_{i=1}^{N} \mathbf{p}_i^{\perp T} \mathbf{F}_i \tag{2.5}$$

Assuming that the object is initially at rest and that the resultant translational friction force and external disturbances are negligible, the object's CoM remains approximately stationary throughout the manipulation process, thereby approximately coinciding with the ICR. Consequently, $\mathbf{p}_C \approx \mathbf{p}_{\text{ICR}}$, and since the CoM remains stationary, it can be treated as a constant parameter to be estimated from measurements collected over time. Multiplying (2.4) by ω_o yields, for robot i at time step k, the

linear measurement model

$$\underbrace{\omega_o(k)\,\mathbf{p}_i(k) + \dot{\mathbf{p}}_i^{\perp}(k)}_{\mathbf{y}_i(k)} = \underbrace{\omega_o(k)\,\mathbf{I}_2}_{\mathbf{\Phi}(k)}\,\mathbf{p}_C + \boldsymbol{\varepsilon}_i(k), \tag{2.6}$$

where $\boldsymbol{\varepsilon}_i(k)$ accounts for measurement noise and modeling errors. Each robot then locally estimates $\mathbf{p}_C$ from (2.6) using standard recursive least squares (RLS) with a forgetting factor $0 < \mu \leq 1$.

The desired torque, whose magnitude should be sufficient to initiate rotational motion, can be agreed upon by all robots through communication or a consensus algorithm. Since the required torque is not known a priori, the trial can be repeated with a progressively increased τ_d, within the feasible torque range of the formation, until rotational motion is observed.

The estimation problem can now be formulated into two sub-problems: formation synthesis and force control. The objective of the formation synthesis problem is to find a configuration $\mathbf{p} = [\mathbf{p}_1^T, \ldots, \mathbf{p}_N^T]^T \in \mathbb{R}^{2N}$ that is able to cooperatively exert the desired force and torque on the object. The force control problem determines the forces to be exerted by each robot to achieve the desired net force and torque on the object.

2.1.3 Formation synthesis

Before deriving a method for determining the robot configuration around the object, it is necessary to first define the set of feasible contact points. Since contacts are restricted to the object boundary, the contact point search space is constructed directly along its polygonal edges. A small margin d_s around each vertex is omitted to ensure that each robot establishes a well-defined contact with a single edge. Denoting the vertices of the polygon by $\mathbf{v}_1, \ldots, \mathbf{v}_M$ (with $\mathbf{v}_{M+1} = \mathbf{v}_1$) and the length of the k-th edge by $\ell_k = \|\mathbf{v}_{k+1} - \mathbf{v}_k\|$, the set of feasible contact points is given by $\mathcal{C} = \bigcup_{k=1}^{M} \left\{ \mathbf{v}_k + s\,\frac{\mathbf{v}_{k+1}-\mathbf{v}_k}{\ell_k} \;\middle|\; s \in [d_s,\, \ell_k - d_s] \right\}$, and the feasible configuration space is defined as $\mathcal{P} = \left\{ \mathbf{p} \in \mathbb{R}^{2N} \mid \mathbf{p}_i \in \mathcal{C},\ i = 1, \ldots, N \right\}$. As for the robot positions $\mathbf{r}_i = \mathbf{p}_i - r_R \mathbf{n}_i$, the edges are dilated outward by r_R to obtain the corresponding edges along which the robot centers can be positioned while maintaining contact with the object. The resulting feasible search space is illustrated in :

Since the admissible force set $\mathcal{F}$ is closed and bounded and the wrench

mapping is continuous, the corresponding manipulation space is a closed set $\mathcal{M}$ consisting of all feasible force–torque combinations, or wrenches, that can be exerted by the current formation. The exact manipulation space cannot be accurately determined or analyzed without prior information about the CoM position, which heavily affects the torque generated by the applied forces. However, under the zero net force condition, this dependence vanishes, allowing the feasible torque range to be determined independently of the CoM position by solving the following linear programming problems:

$$\begin{aligned} \tau_{\max}(\mathbf{p}) &= \max_{\mathbf{f}\in\mathcal{F}} \sum_{i=1}^{N} \mathbf{p}_i^{\perp T}\mathbf{F}_i \quad \text{s.t.} \quad \sum_{i=1}^{N} \mathbf{F}_i = \mathbf{0}, \\ \tau_{\min}(\mathbf{p}) &= \min_{\mathbf{f}\in\mathcal{F}} \sum_{i=1}^{N} \mathbf{p}_i^{\perp T}\mathbf{F}_i \quad \text{s.t.} \quad \sum_{i=1}^{N} \mathbf{F}_i = \mathbf{0}. \end{aligned} \tag{2.7}$$

These two quantities define the endpoints of the interval obtained by intersecting the manipulation space with the torque axis. A formation is considered feasible if the desired torque τ_d is within the range $[\tau_{\min}(\mathbf{p}), \tau_{\max}(\mathbf{p})]$. It is also beneficial to maximize the minimum distance between the desired torque τ_d and the boundary of the feasible torque range, which can be formulated as a robustness metric J_r:

$$J_r(\mathbf{p}) = \min\left(\tau_{\max}(\mathbf{p}) - \tau_d,\ \tau_d - \tau_{\min}(\mathbf{p})\right). \tag{2.8}$$

The formation synthesis problem is formulated as:

$$\min_{\mathbf{p}\in\mathcal{P}} \quad J(\mathbf{p}) = -J_r(\mathbf{p}) \tag{2.9a}$$

$$\text{s.t.} \quad \exists\, \tilde{\mathbf{f}} \in \mathcal{F} \quad \text{s.t.} \quad \sum_{i=1}^{N} \tilde{f}_i \mathbf{n}_i = \mathbf{0},\ \sum_{i=1}^{N} \mathbf{p}_i^{\perp T}(\tilde{f}_i \mathbf{n}_i) = \tau_d, \tag{2.9b}$$

$$\|\mathbf{r}_i - \mathbf{r}_j\| \geq d_{\min}, \qquad \forall i \neq j. \tag{2.9c}$$

where $d_{\min} > 2r_R$ is the minimum Euclidean distance between the robots to avoid collision. It is essential that the constraints (2.9b) and (2.9c) are satisfied to ensure that the desired wrench can be exerted by the formation without violating physical feasibility, while minimizing the objective function yields a configuration that is optimal with respect to the selected criteria. The objective function (2.9a)

encodes a desirable property, which can be modified to include other metrics. In this case, only rotational robustness is considered. The formation synthesis problem is a non-convex optimization problem, which can be addressed using heuristic approaches such as particle swarm optimization (PSO). To maintain the distributed nature of the system, each robot can run the optimization algorithm independently and share its best particle with its neighbors. Once the optimal contact configuration has been determined, the robots autonomously negotiate which contact position each robot should occupy. Based on the resulting assignment, each robot then navigates toward its designated contact point on the object boundary, thereby allowing the team to establish the desired manipulation configuration in a distributed manner.

2.1.4 Force control

The next problem to be solved is to determine the forces to be exerted by each robot to achieve the desired net force and torque on the object. While one naive approach would be to simply solve (2.1) and (2.3) using a least-squares approach, this would fail to account for the physical limitations of the forces exerted by the robots, potentially yielding a physically infeasible solution. To address this limitation, the force control problem is formulated as an optimization problem:

$$\min_{\mathbf{f}} \quad \left(\sum_{i=1}^{N} \mathbf{p}_i^{\perp T}\mathbf{F}_i - \tau_d\right)^2 + \lambda\|\mathbf{f}\|_2^2, \qquad \lambda > 0 \tag{2.10}$$

$$\text{s.t.} \quad \mathbf{F} = \mathbf{0}, \tag{2.11}$$

$$0 \le f_i \le f_{\max}, \qquad i = 1, \ldots, N. \tag{2.12}$$

Satisfying constraints (2.11) and (2.12) is essential to ensure force balance and compliance with actuator limits, while the objective function encourages the generation of the desired rotational torque. It is desirable that the net torque be as close as possible to the desired torque τ_d; however, it is not required to exactly match τ_d since only a torque sufficient to induce rotational motion is required for the estimation task. The second term in the objective function (2.10) is a quadratic regularization term that penalizes large force magnitudes, which is beneficial for energy efficiency and avoiding actuator saturation. The regularization coefficient λ is chosen to be sufficiently small to prioritize torque tracking while discouraging

unnecessarily large pushing forces. The optimization problem can be solved using standard convex optimization techniques, such as quadratic programming. The uniqueness of the optimal solution can be established by examining the convexity of the objective function. Let $\mathbf{a} = [a_1, \ldots, a_N]^T$, where $a_i = \mathbf{p}_i^{\perp T}\mathbf{n}_i$. The Hessian matrix of the objective function (2.10) is given by $\mathbf{H} = 2\mathbf{a}\mathbf{a}^T + 2\lambda\mathbf{I}$. Since $\mathbf{a}\mathbf{a}^T$ is positive semidefinite and $\lambda > 0$, the Hessian is positive definite, implying that the objective function is strongly convex. Furthermore, the zero net force constraint and actuator limits define a nonempty, compact, and convex feasible set. Consequently, the optimization problem admits a unique global optimal solution. Each robot can independently solve this optimization problem using the formation obtained above, ensuring a distributed implementation of the force control strategy.

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